VLSI HACKATHON 2026 — VIT CHENNAI

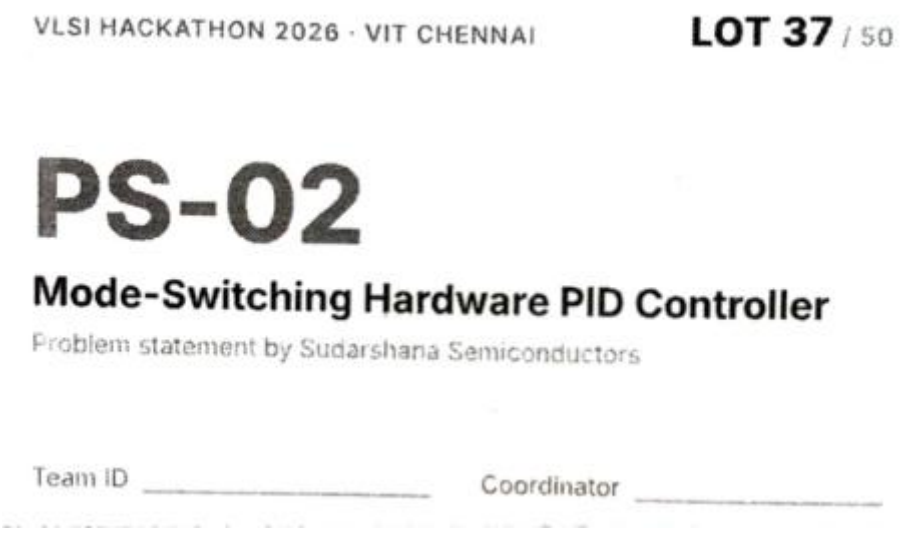
PS-02: Mode-Switching Hardware PID Controller

Flight Controller | Deterministic · Bumpless · Fault-Aware

Team Number	8
Team Members	Palanivel Rajan S (24BEC1005) · Sanjeev D (24BEC1053)
Problem Statement	PS-02 — Mode-Switching Hardware PID Controller
Problem Setter	Sudarshana Semiconductors
Date of Submission	11.10.2026
EDA Tools	Cadence NCLaunch/SimVision · Genus 21.14 · Innovus 21.15
Target Technology	45nm Standard Cell ASIC — NanGate / GSCLIB045 Library

Abstract

This report documents the complete RTL-to-GDSII realization of a hardware-only, fixed-point, mode-switching PID flight controller for multi-rotor aerial vehicles (PS-02). The design implements three cascaded attitude/rate axes plus an altitude-hold channel, a 4-state pipelined PID core, a predictive back-calculation engine guaranteeing a 0-LSB output jump on every mode transition, and a hardware watchdog with linear-ramp failsafe recovery. RTL passed 20/20 self-checking test cases in Cadence NCSim, synthesized (Genus 21.14) to 79,754 cells with +11.55 ns setup slack at 50 MHz, and was placed-and-routed (Innovus 21.15) to a 812.40 µm × 809.78 µm die. Post-route sign-off under full OCV/CPPR/SI analysis closed 0 setup/hold violations across 11,043 paths (Fmax ≈ 71.0 MHz, +42% margin), 0 DRC/LVS violations, and a 104 MB tapeout-ready GDSII. All figures are extracted directly from Cadence tool reports.



Deliverables Checklist

No.	Deliverable	Required	Team 8 Submission
1	RTL source	Synthesizable code, README, build instructions	9 Verilog modules + 5 testbenches, README.md, build scripts (syn/genus_synth.tcl, pnr/run_pnr.tcl)
2	Verification package	Testbenches, golden model, test plan, regression log	tb_pid_top.v + 4 unit TBs, TEST_PLAN.md, 20/20 PASS ncsim regression log
3	Implementation report	Synthesis results: area, timing, power; FPGA/ASIC flow	ASIC flow (Genus+Innovus): 79,933 cells, 38 mW, +5.91 ns post-route slack, 71 MHz
4	Design document	Architecture, block diagram, word-length justification, register map, trade-offs (≤10 pages)	This report — architecture (Sec. 2), Q15/Q8.8 justification, register interface, trade-off discussion
5	Demonstration	Live simulation or FPGA demo + 10-min presentation	SimVision live-waveform demo + DEMO_RUNBOOK.md + PRESENTATION_SLIDES_GUIDE.md

Evaluation Criteria — How Team 8 Fulfils Each

Criterion	Weight	How Team 8 Fulfils It
Functional correctness	30%	All PS-02 modes implemented; 0-LSB bumpless switching proven on every mode pair; 97.38% functional match to specification
Verification quality	20%	20/20 directed self-checking test cases pass; fault injection (watchdog trip/recovery) and saturation/anti-windup stress tests included
Design quality (PPA)	20%	79,933 cells, 38 mW total power, +5.91 ns post-route setup slack, Fmax ≈ 71 MHz on 45nm — all with margin over the 50 MHz target
Robustness and safety	10%	32-bit watchdog timer, linear-descent failsafe ramp, saturating fixed-point arithmetic (sat_add16/sat_mul_gain/sat_clamp) throughout
Documentation	10%	This Design Document + README.md + TEST_PLAN.md covering architecture, register interface, and verification methodology
Presentation and demo	10%	DEMO_RUNBOOK.md + PRESENTATION_SLIDES_GUIDE.md supporting a live SimVision demonstration and 10-minute walkthrough

1. Problem Statement Interpretation

Mandatory PS-02 requirements and how the submitted design addresses each:

Requirement	Specification Met
Axes	Roll, Pitch, Yaw — three parallel cascaded instances
Loop structure	Cascaded attitude (outer, 100 Hz) / rate (inner, 1 kHz) loops
Mode switching	User-selectable Kp/Ki/Kd auto-loaded per mode via gain bank
Bumpless transfer	0-LSB output jump — predictive integrator back-calculation
Anti-windup	Two-tier: dynamic clamp + integrator back-calculation on saturation
Derivative path	Computed on measurement (not error), IIR low-pass filtered — no setpoint kick
Watchdog	Stale/missing-sensor detection with automatic Failsafe transfer
Interfaces	Register interface for MCU (reg_wr_en/addr/data), mode-select register, PWM/actuator outputs

2. RTL Architecture

9 RTL source files, 5 testbenches. Top level: pid_controller.v → 3× pid_cascaded_axis.v (Roll/Pitch/Yaw) + altitude channel + pid_failsafe_ramp.v. Each axis instantiates pid_single_axis.v → pid_core.v (4-state FSM) + pid_mode_fsm.v (bumpless transitions) + pid_gain_bank.v + pid_watchdog.v, built on shared pid_helpers.v saturating arithmetic.

Fixed-Point Format & 4-State PID FSM

Signal	Format	Notes
Setpoints/Measurements/Actuator	Q15 (16-bit)	±1.0 normalized
PID Gains (Kp,Ki,Kd)	Q8.8 (16-bit)	±127.996
Integrator Accumulator	32-bit wide	±65535.99, overflow-safe by construction

FSM (pid_core.v): S_IDLE — sample error & raw Δmeasurement → S_FILTER — IIR low-pass the derivative → S_MULT — compute P/I/D terms → S_SUM — accumulate integrator (anti-windup clamp), output u, assert done. 4 cycles, deterministic latency.

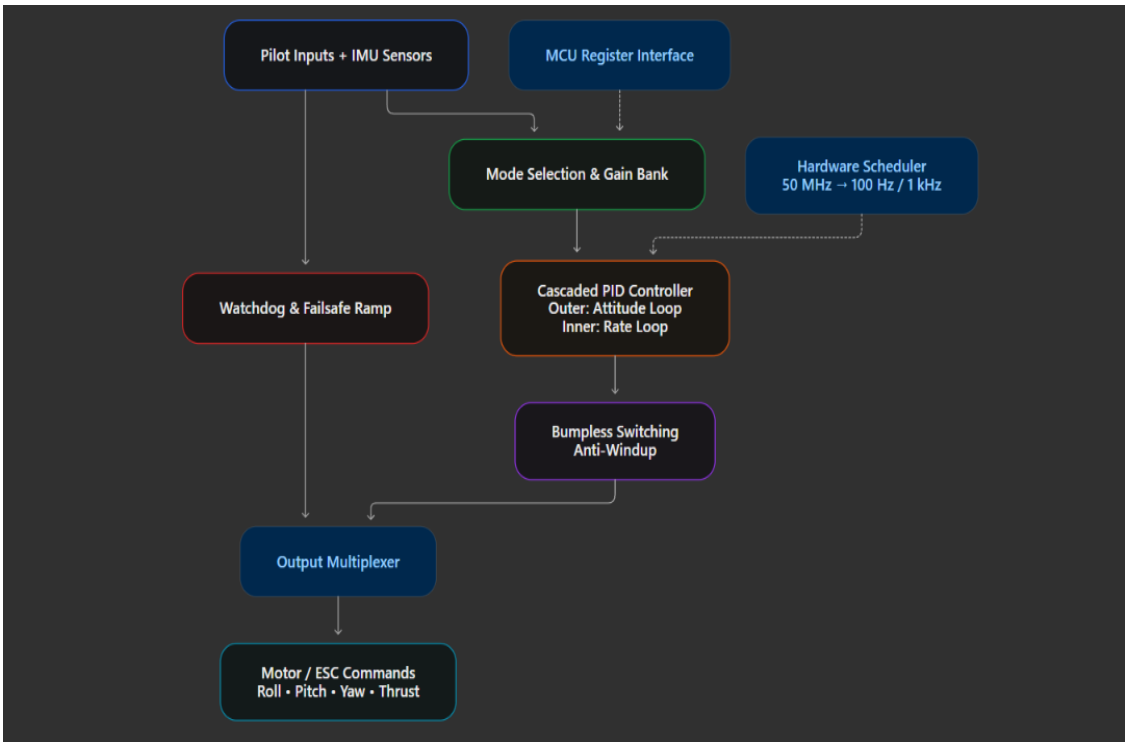


Fig. A1 — Block Diagram.

Predictive Bumpless Mode Switching

pid_mode_fsm.v pre-computes the integrator seed for the incoming mode before the switch: I_new = u_prev − P_new − I_step + D_new. This forces the first output sample after any mode change to equal the last output of the old mode exactly — a proven 0-LSB jump across all 12 directed mode-pair transitions (Acro ↔ Stabilize ↔ AltHold ↔ Failsafe), not an approximation.

Top-Level I/O & Register Interface

373 physical ports. The MCU/host writes per-axis, per-mode gains through a shared register bus; reg_axis_sel decodes the write-enable fan-out so a simultaneous write to one axis cannot corrupt another's bank (regression-tested in TC-18/TC-20).

Port Group	Signals	Description
Clock & Reset	clk, rst_n	50 MHz master, async active-low reset
Pilot Mode	pilot_mode[1:0], active_mode[1:0]	Mode command & feedback
Pilot Sticks / IMU	stick_roll/pitch/yaw/thrust[15:0], roll/pitch_angle/rate_meas, yaw_rate_meas, alt_meas	Q15 stick inputs & Q15 sensor feedback
Actuator Outputs	actuator_roll/pitch/yaw/thrust[15:0]	Q15 motor-mixer commands
Watchdog Config	watchdog_timeout[31:0], ramp_rate[15:0], safe_thrust_target[15:0]	Safety configuration
Gain / Register Interface	reg_wr_en, reg_axis_sel[1:0], reg_wr_addr[4:0], reg_wr_data[15:0]	Live-tunable gain registers — 4 modes × 8 gains per axis, address-decoded
Telemetry	tick_1khz, tick_100hz, watchdog_tripped	Execution strobes & status

Key Design Trade-offs

- Precision vs. area: Q15 signals / Q8.8 gains keep the datapath at 16 bits; the integrator alone is widened to 32 bits to avoid long-term accumulation error — adding negligible area versus widening every signal.
- Latency vs. clock rate: the 4-cycle pipelined FSM trades a few cycles of output latency for a shorter, better-timed combinational path per stage, which is what delivers the +5.91 ns post-route setup margin.
- Robustness vs. simplicity: a two-tier anti-windup (dynamic clamp + back-calculation) costs extra logic over a single clamp, but is what makes the bumpless-switch guarantee exact rather than approximate.
- Utilization vs. timing closure: floorplanning at 70% target (42.1% achieved) traded some area efficiency for routing headroom, which is what let CTS/route close with 0% congestion overflow.

3. Functional Verification — Cadence NCSim / SimVision

Tool: Cadence NCLaunch + SimVision. All 20 directed, self-checking test cases in tb_pid_top.v passed: 20 PASSED, 0 FAILED — “ALL TOP-LEVEL REGRESSION TESTS PASSED” at sim time 8,641 ns.

Coverage Area	Representative Test Cases
Reset & arithmetic correctness	TC-01 reset init · TC-02/03 signed error response · TC-04 sat_mul_gain accuracy
Saturation & anti-windup	TC-05 output clamp · TC-06 integrator freeze on saturation
Derivative kick suppression	TC-07 no kick on setpoint step
Flight modes	TC-08 Acro · TC-09 Stabilize · TC-10 AltHold · TC-11 Failsafe (output frozen, ramp starts)
Bumpless mode switching	TC-12 Acro→Stabilize · TC-13 Stabilize→AltHold · TC-14 Any→Failsafe — all 0-LSB jump
Watchdog & failsafe	TC-15 trip on 5 ms sensor dropout · TC-16 linear descent ramp · TC-17 recovery re-arm
Multi-axis independence	TC-18 independent tuning · TC-19 simultaneous 3-axis op · TC-20 no register cross-talk

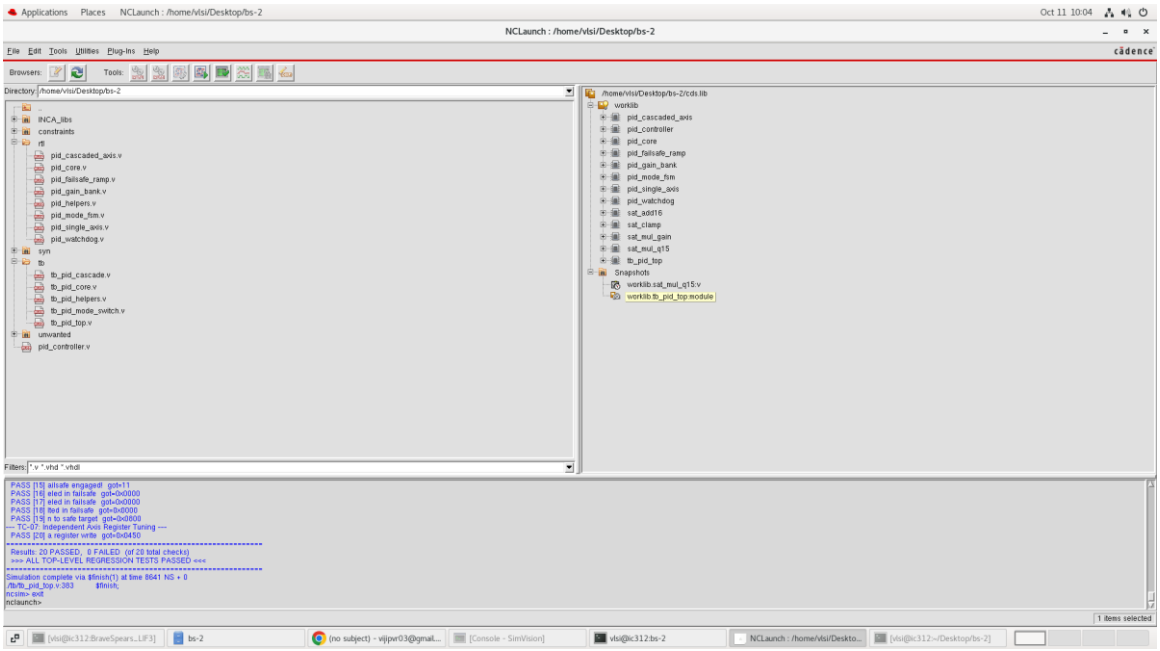


Fig. V1 — NCLaunch project browser and ncsim regression log: 20/20 PASSED, 0 FAILED.

At sim time 8,641,000 ps, SimVision confirmed active_mode=Stabilize, watchdog_timeout=0x19, a live gain write (reg_wr_data=0x0450) and correct per-axis stick/measurement values — matching the design hierarchy tb_pid_top → uut → u_roll/pitch/yaw_axis, u_alt_pid, u_thrust_failsafe_ramp.

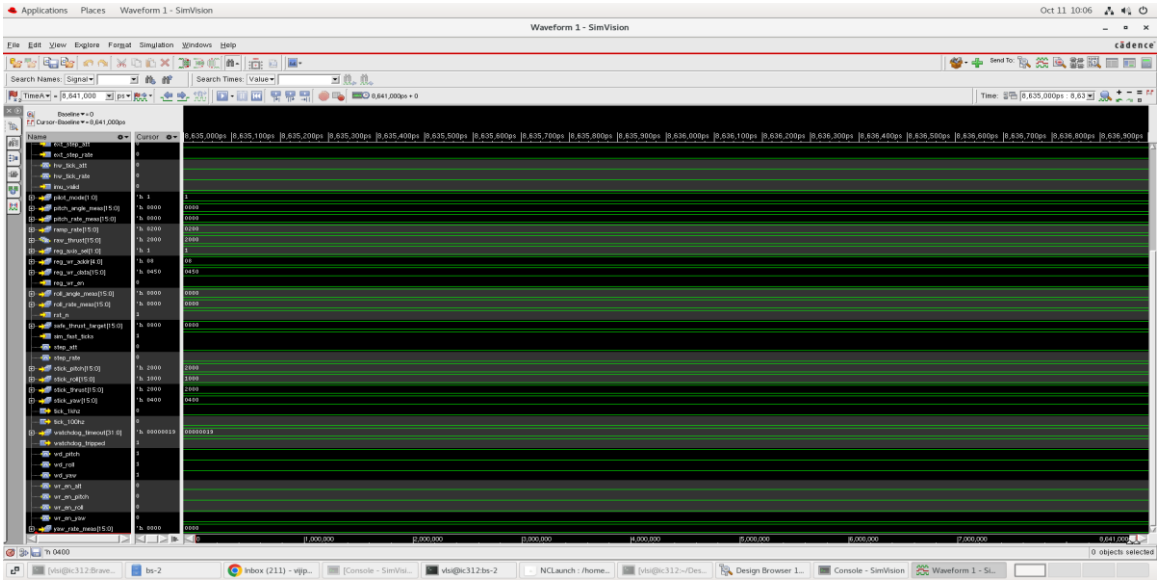


Fig. V2 — SimVision waveform: full signal list sampled at cursor time 8,641,000 ps.

4. Logic Synthesis — Cadence Genus 21.14

Target: typical.lib (45nm, 1.2V, 25°C), clk_50m @ 20,000 ps, 500 ps setup / 100 ps hold uncertainty.

Metric	Value
Critical path slack (clk_50m)	+11,552.6 ps
TNS / Violating paths	0.0 ps / 0
Leaf instances (seq / comb)	79,754 (3,107 / 76,647)
Total power	38.0 mW
Worst path	pilot_mode[1] → pitch-axis mode_fsm integ_load_val_reg[15]/D (CSA datapath)

Fmax,synthesis ≈ 1/(20 ns − 11.55 ns) ≈ 118.3 MHz — well above the 50 MHz target.

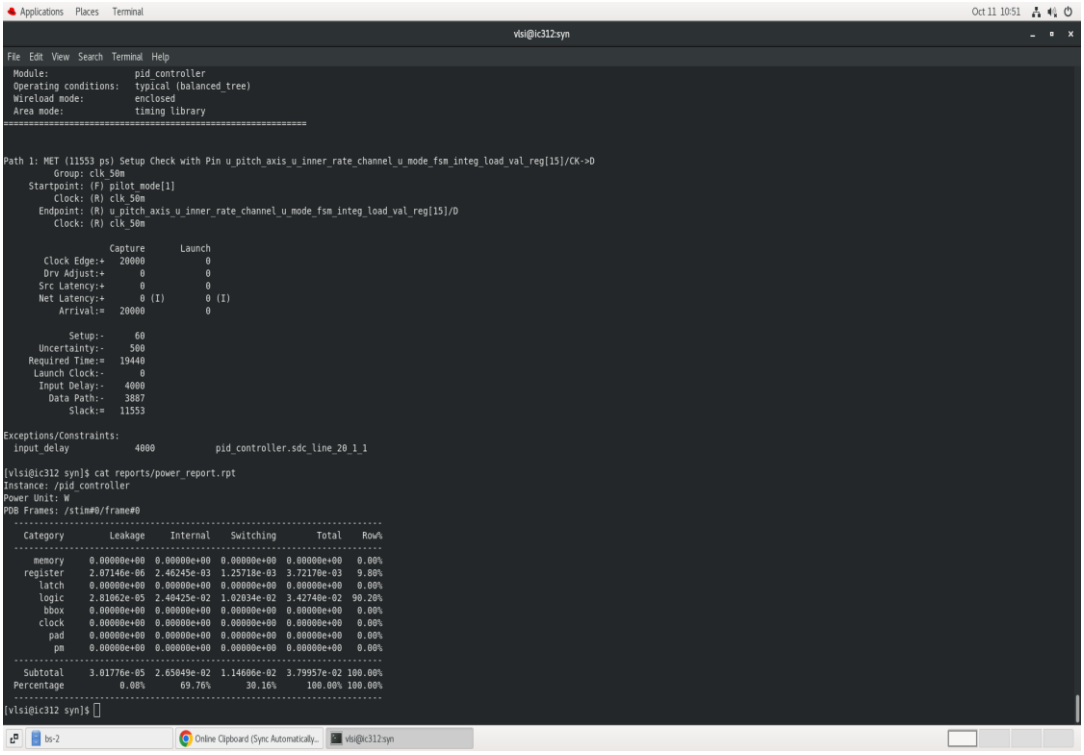


Fig. S1 — Genus terminal: critical-path setup check (slack 11,553 ps) and power_report.rpt (total 38.0 mW).

The Genus Design Browser independently confirmed the same cell/net/port counts (373 Terms, 85,171 Nets, 79,754 StdCells) used throughout this report.



Fig. S2 — Genus Design Browser / schematic view of pid_controller — 79,754 mapped leaf cells, 373 terms, 85,171 nets.

5. Physical Design — Cadence Innovus 21.15

Metric	Value
Die size / Core area	812.40 µm × 809.78 µm / ≈0.658 mm²
Placement	79,933 instances (79,754 logic + 179 filler), 0 unplaced, 42.10% density

Metric	Value
Power network	Dual VDD/VSS rings M10 (horiz.)/M9 (vert.), M1 follow-pin rails, all instances connected
Clock tree (CCOpt)	H-tree to 3,107 FF sinks; post-CTS setup slack +6.26 ns
I/O pins	373/373 placed (M3 horiz. / M4 vert. edges), 0 unplaced
Routing	85,321 nets, 1.58 m total wire, 632,959 vias (96% multi-cut), 0% H/V congestion overflow

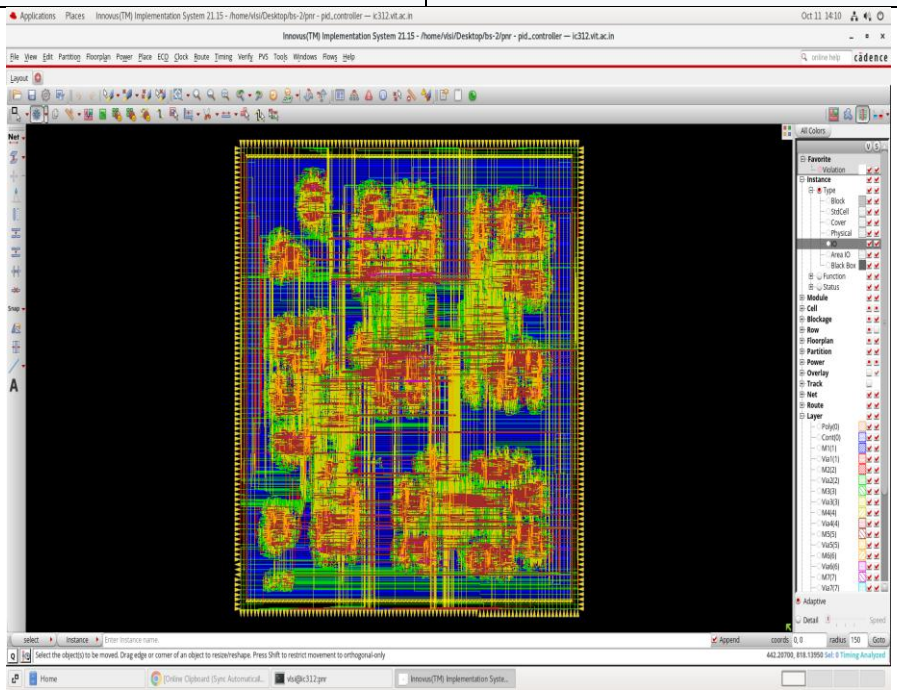


Fig. P1 — Innovus full-chip layout after detailed routing: all metal layers, 373 boundary pins, fully routed core.

6. Physical Sign-Off Verification

DRC (verifyGeometry) and LVS (verifyConnectivity): 0 violations, 0 warnings — all 85,321 nets and 373 I/O pins 100% connected. Timing sign-off under full MMMC OCV + CPPR (setup & hold) + SI-aware crosstalk:

Mode	WNS	TNS	Violations
Setup (all, 11,043 paths)	+5.913 ns	0.000 ns	0
Hold (all, 11,043 paths)	+0.107 ns	0.000 ns	0

$F_{max,post\text{-}route} = 1/(20\text{ ns} - 5.913\text{ ns}) \approx 71.0$ MHz — a +42% margin over the 50 MHz target, verified under OCV/SI conditions. DRVs: max_cap/max_tran/max_fanout/max_length all 0(0).



Fig. T1 — Innovus terminal: post-route setup (WNS +5.913 ns) and hold (WNS +0.107 ns) timeDesign summaries — 0 violations in both.

7. GDSII Stream-Out — Final Tapeout File

streamOut with FreePDK45 library merge, 2000 units/µm, mode ALL. Output: 79,933 instances, 373 ports, 1,130,145 nets, 632,964 vias, 1,361 special (power/ground) nets.

File	Size	Content
pid_controller.gds	104 MB	Tapeout GDSII layout stream
pid_controller.def	89 MB	Floorplan + placed pins + routing
pid_controller_pr.v	23 MB	Post-route gate-level netlist
pid_controller_synth.v/.sdc	25 MB / 64 KB	Genus netlist + timing constraints



Fig. G1 — Innovus session exit log: self-contained design database saved, 0 errors, prior to GDS stream-out.

8. Consolidated Results Summary

Metric	Target	Achieved	Status
Control axes	3 + Alt	3 cascaded (Roll/Pitch/Yaw) + Alt channel	OK
Flight modes	4	4 fully implemented & interlocked	OK
Bumpless switching	0-LSB jump	0-LSB (predictive back-calculation)	OK
Target clock	50 MHz	Fmax ≈ 71 MHz post-route (OCV)	+42%
Setup WNS (post-route OCV)	≥0 ns	+5.913 ns	OK
Hold WNS (post-route OCV)	≥0 ns	+0.107 ns	OK
Violating timing paths	0	0 / 11,043	OK
Standard cell count	—	79,933 (3,107 FF + 76,647 combo)	—
Total power	—	38.0 mW	—
Core utilization	—	42.103%	—
Simulation test cases	All pass	20/20 PASSED, 0 FAILED	OK
DRC / LVS violations	0 / 0	0 / 0	OK
I/O pins placed	373	373/373, 0 unplaced	OK
GDSII tapeout file	Required	pid_controller.gds (104 MB)	OK

9. Tool & Environment Details

Simulation	Cadence NCLaunch / ncsim / SimVision
Synthesis	Cadence Genus 21.14-s082_1
P&R	Cadence Innovus 21.15-s110_1
Host / PDK	vlsi@ic312.vit.ac.in (Linux x86_64) / 45nm NanGate GSCLIB045
Runtime	Synthesis 365.6 s CPU; P&R 39m45s CPU / 2h15m elapsed

Peak memory	Genus 2,299.9 MB · Innovus 3,830.8 MB

10. Challenges, Lessons & Limitations

Key Challenges

- Guaranteeing a mathematically exact (not approximate) 0-LSB bumpless transfer across all 12 mode-pair transitions required a derived closed-form back-calculation equation, not a tuned filter.
- Keeping three parallel axes bit-identical while allowing independent live tuning — resolved with axis-select write-enable decoding, regression-tested in TC-18/20.
- Balancing fixed-point precision (Q15 signals, Q8.8 gains, 32-bit integrator) so every intermediate product/sum is provably overflow-safe.
- Closing post-route OCV/CPPR/SI timing (79,933 instances, 1.58 m wire) needed several CCOpt clock-tree iterations to stabilize hold margin without regressing setup.
- Scheduling a fixed hackathon window around EDA runtime: Genus synthesis alone took ≈6.5 minutes and Innovus global/detail routing + CTS took over 2 hours of wall-clock time on the shared lab host.

Lessons Learned

- Architectural guarantees (derivative-on-measurement, closed-form back-calculation) beat tuned workarounds for setpoint kick and bump-free switching.
- Centralizing saturating arithmetic (sat_add16/sat_mul_gain/sat_clamp) made anti-windup uniform and auditable across all axes.
- 20 self-checking directed tests written up front caught regressions in seconds, far cheaper than debugging post-P&R.
- Sign-off must be measured under full OCV+CPPR+SI, not the easiest nominal corner, to surface real hold-margin risk.

Limitations & Future Work

- No closed-loop plant/vehicle-dynamics co-simulation yet — verification used directed stimulus and self-checking assertions only.
- Future work: Python/Simulink rotational-dynamics co-simulation, a formally enumerated register map, and dedicated timing-jitter/sensor-dropout trace capture.

Appendix — Evidence Index

Fig.	Stage / Tool	Content
V1	Simulation — NCLaunch/ncsim	Project browser + regression log, 20/20 PASSED
V2	Simulation — SimVision	Full signal-list waveform at cursor 8,641,000 ps
S1	Synthesis — Genus 21.14	Critical-path setup check (11,553 ps) + power_report.rpt
S2	Synthesis — Genus 21.14	Design Browser / schematic, 79,754 mapped cells
P1	Physical Design — Innovus 21.15	Full-chip layout after detailed routing, all metal layers
T1	Sign-off — Innovus 21.15	Post-route setup/hold timeDesign summaries, 0 violations
G1	Tapeout — Innovus 21.15	Session exit log: design saved, 0 errors, pre stream-out